Interactive electrothermal design of Joule-heated reactors

Yeonsu Kwak*

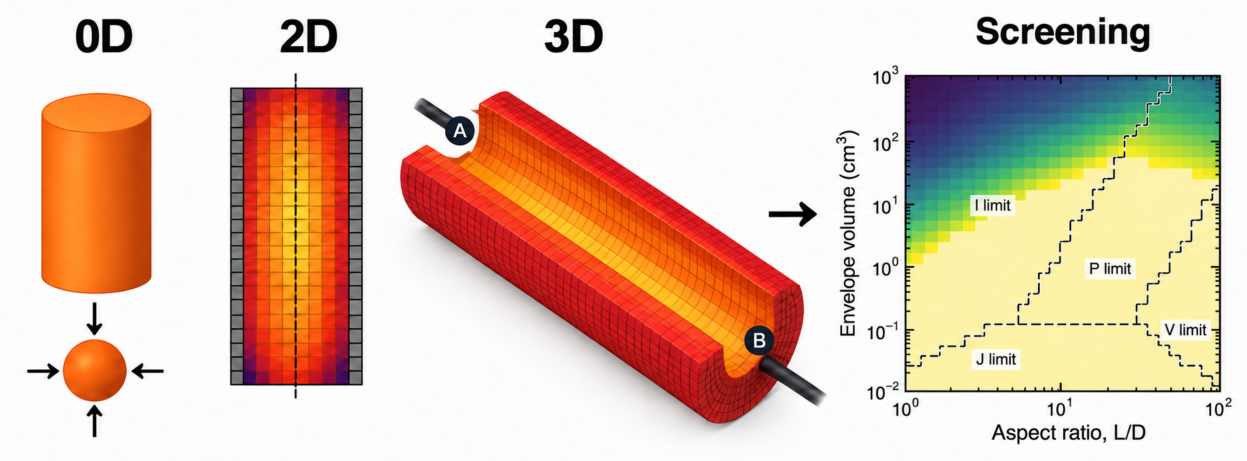
Department of Chemical and Biomolecular Engineering,
University of Delaware, Newark, Delaware 19716, United States

*E-mail: fiske@udel.edu

Abstract

Designing Joule-heated reactors requires matching element resistance and thermal losses to the available power supply. We present ScreenJoule, an open-source browser toolkit combining lumped, axisymmetric, and three-dimensional electrothermal models to evaluate materials, element dimensions, and electrode configurations. The calculations show that wider electrode contacts improve temperature uniformity at fixed power but can reduce delivered power when the supply reaches its current limit. Elements with the same material and similar solid volume develop different mean temperatures and hot spots at equal power because their shapes alter exposed area, current distribution, and heat-conduction paths. Parameter sweeps identify combinations of element size, aspect ratio, and supply ratings that reach a target temperature and quantify their startup times. Analytical benchmarks, mesh refinement, and matched OpenFOAM calculations verify the numerical implementation. ScreenJoule provides an interactive framework for evaluating electrode design and reactor-element scale-up under practical electrical and thermal constraints.

Graphical Abstract



1. Introduction

Direct resistance heating generates heat within an electrically conducting reactor element, shortening the heat-transfer path associated with furnace heating.[1,2] Electrified methane reforming, structured SiSiC reactors, and internally heated hydrocarbon crackers demonstrate its use in high-temperature chemical processing.[3–7] An experimentalist must select the element dimensions and mass, determine whether the available supply can reach the target temperature, and assess heating rate and temperature variation within the element.

These choices are linked through electrical resistance and the heat balance. Changing an element's length or cross-section changes its resistance, thermal mass, and exposed surface area. At fixed voltage, the resistance sets Joule power until a current or power limit is reached. A change in aspect ratio can therefore shift operation between current-limited and voltage-limited regimes.[1,7] The net power after heat loss, divided by the element's heat capacity, determines the lumped heating rate; the steady temperature is reached when heat generation balances heat loss. Temperature-dependent resistivity further changes the operating point during heating. Electrode contact geometry and thermal conduction determine how the resulting heat is distributed, so an adequate mean temperature can coexist with local overheating.

Lumped energy balances and spatially resolved models describe different parts of this design problem. Mittal et al. combined three-dimensional Joule-heating CFD with a lumped energy-balance model,[8] while Sundaramoorthy et al. compared volume-averaged heat-transfer models for structured reactors with channel-resolved CFD.[9] Scientific software also makes calculations accessible for repeated use: Energy2D provides interactive heat-transfer simulations,[10] fuelcell combines electrochemical data processing with a graphical interface,[11] and OpenMKM provides reusable multiscale catalytic reactor calculations.[12] For initial Joule-heater design, the practical task is to evaluate material, dimension, and supply combinations across a broad range, then resolve current paths and temperature gradients in selected configurations.

We use ScreenJoule to examine reactor-element scale-up and electrode design, quantifying how material, dimensions, electrode contact area and placement, and supply limits affect heating capacity, startup time, and temperature uniformity. Wider electrode contacts improve uniformity at fixed power but reduce available heating once the current ceiling is reached. The models treat rods, tubes, tapes, foams, and monolith walls with a continuous conducting path between electrodes; solid fraction and effective properties represent porous elements. Particle beds require a separate contact-network description because compression and particle contacts determine their resistance and local heating.

2. Model Formulation

2.1. Software workflow

Joule0D determines the resistance, available power and lumped temperature for a selected material, geometry and power supply. Joule2D resolves axisymmetric element and enclosure temperatures, including heating and cooling transients (**Figure 1**). Joule3D solves the electrical potential and temperature in a three-dimensional solid and screens pairs of design variables (**Figure 4**). Comparisons between workspaces require matched conditions. Calculated data can be exported for analysis. Figure 1c–g summarizes the lumped and axisymmetric formulations.

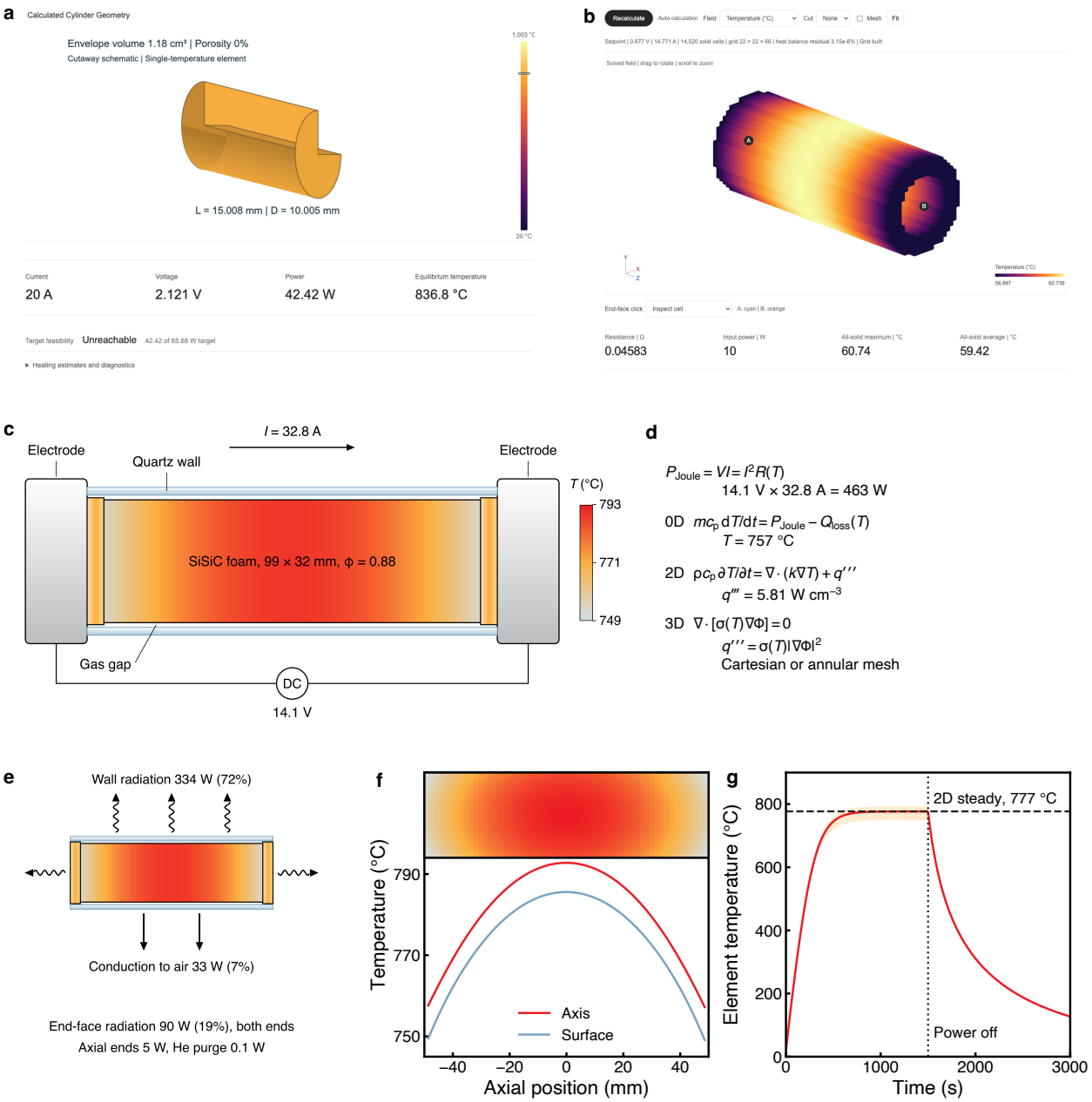


Figure 1. ScreenJoule workspaces and a solved electrothermal case. **(a)** Lumped (0D) operating point of a dense SiC cylinder. **(b)** Three-dimensional temperature field of a hollow tube at 10 W, with electrode locations A and B. **(c–g)** Zheng SiSiC foam (99 × 32 mm, $\phi = 0.88$) at 14.1 V: **(c)** element in a quartz tube, colored by the solved axial temperature, with gap and wall drawn wider than scale, **(d)** governing relations of the 0D, 2D and 3D models, **(e)** boundary heat-loss channels, which sum to the 463 W terminal power, **(f)** sectional temperature field (top) with axis and surface profiles, and **(g)** mean element temperature (line) and its range (band) during 1500 s of heating and 1500 s of cooling.

2.2. Lumped and axisymmetric operating points

For a cylindrical element with solid inventory V_s, void fraction ϕ, and aspect ratio $\lambda = L/D$, the solid fraction is $f_s = 1 - \phi$ and the envelope volume is $V_{env} = V_s / f_s$. The dimensions and effective

conducting area are

$$\begin{aligned} D &= \left(\frac{4V_{env}}{\pi\lambda}\right)^{1/3} \\ L &= \lambda D \\ A_{eff} &= f_s \frac{\pi D^2}{4} \end{aligned} \tag{1}$$

The lumped model evaluates element resistance at its uniform temperature. The axisymmetric model uses the volume-average element temperature and includes two equal contact resistances,

$$\begin{aligned} R_{bulk} &= \rho_e(T_{avg}) \frac{L}{A_{eff}} \\ R_{total} &= R_{bulk} + 2R_c \end{aligned} \tag{2}$$

The lumped calculation uses $R_{total} = R_{bulk}$. Voltage, current and power ratings apply simultaneously: a 150 V / 40 A / 2 kW supply can deliver at most 13.3 A at 150 V. In automatic mode, the lumped and axisymmetric solvers select the current from the most restrictive of four ceilings,

$$I = \min\left(I_{\max}, \quad \frac{V_{\max}}{R_{total}}, \quad \sqrt{\frac{P_{\max}}{R_{total}}}, \quad J_{\max} A_{eff}\right) \tag{3}$$

and deposit $P_{bulk} = I^2 R_{bulk}$ in the element. The axisymmetric model reports contact dissipation separately; its thermal source is the bulk element power.

2.3. Axisymmetric power distribution

A local electrical calculation determines where the bulk power is deposited (Figure 1c). Within the element the electric potential satisfies

$$\nabla \cdot [\sigma(T) \nabla \Phi] = 0 \tag{4}$$

for a unit potential difference between its axial ends. The local dissipation $q'''_{unit} = \sigma(T)|\nabla\Phi|^2$ is normalized to the scalar bulk power,

$$\begin{aligned} q'''(r,z) &= P_{bulk} \quad \frac{q'''_{unit}(r,z)}{\int_{V_e} q'''_{unit} \; dV} \\ \int_{V_e} q''' \; dV &= P_{bulk} \end{aligned} \tag{5}$$

At each thermal iteration, the mean-temperature circuit calculation sets P_{bulk}. The model distributes this power uniformly over the element volume or, with the local electrical option enabled, according to eq 5.

2.4. Axisymmetric temperature field

The thermal calculation uses an axisymmetric finite-volume domain containing the element, gas gap, containment wall, surrounding air, and axial process-gas regions (Figure 1c). The stationary temperature satisfies

$$\frac{1}{r}\frac{\partial}{\partial r}(kr\frac{\partial T}{\partial r}) + \frac{\partial}{\partial z}(k\frac{\partial T}{\partial z}) + q''' = 0 \tag{6}$$

The model includes conduction across material interfaces, gas-gap conduction, element-to-wall gray-body radiation, wall and surrounding-air heat transfer, boundary convection and radiation, and prescribed process-gas sensible-enthalpy transport. Transient calculations add thermal storage. Temperature-dependent properties and radiative heat transfer are updated iteratively; SI Section S1 gives the discretization, property models and solution procedure.[13,14]

For the opt-in porous-body closure, volumetric element storage is $(1 - \phi)\rho$s cp, where ρs is skeleton density. Solid fraction scales mass and conducting area; effective thermal conductivity is specified independently of this scaling. The present design calculations assume isotropic effective k = 60 W m^{-1} K^{-1} and examine 20 and 120 W m^{-1} K^{-1} separately.

2.5. Three-dimensional electrothermal model

Joule3D updates each cell's resistivity from its temperature and solves the unit-potential problem. The resulting terminal conductance determines resistance and the voltage needed for the requested voltage, current or power setting. Voltage, current and power ceilings constrain the applied voltage. The solver then calculates local Joule heating and updates the temperature field, repeating these steps to convergence. A specified peak-current-density limit stops the calculation when exceeded. Finite electrode regions and electrical contact resistance can be included; contact dissipation enters the adjacent thermal cells.

The optional gas mode solves incompressible Stokes flow and gas energy transport coupled to solid conduction. An invariant channel cross-section uses a fully developed flow solution.

The electrode-width study in **Figure 4** uses steady solid conduction with ideal end contacts. Its thermal boundaries and electrical settings are given with the figure. For the Zheng comparison in Figure S6, a prescribed process heat duty is distributed over the heater volume and subtracted from the local energy balance.

2.6. Numerical verification

For the axisymmetric formulation, the manufactured-solution error decreases across seven meshes (Figure S1A). The doubling sequence 30→60→120 gives L2 spatial orders of 2.05 and 2.06. Five physical meshes resolve the selected porous SiC case (Figure S1B); the peak changes by 0.079 K between the two finest meshes. Radial and annular analytic-limit comparisons, and transient time-step checks, are given in Figure S1. These checks assess numerical implementation.[15]

Six cases compare fully developed gas flow and conjugate heat transfer with an independent OpenFOAM implementation at three flow rates on two matched meshes (Figure S7). Maximum solid and gas temperature differences are 1.26×10^{-6} and 1.05×10^{-6} K, respectively, for matching discrete equations. SI Section S5 reports velocity, pressure drop, energy closure and mesh sensitivity.

Literature comparisons quantify the sensitivity of calculated temperatures to external heat loss and process heat demand (Figure S6).

3. Results and Discussion

3.1. Porous SiC design case

The spatial reference case uses an 80 cm^3 envelope, porosity 0.5 and L/D = 30. Its solid mass is 128.4 g, with a length of 450.9 mm and diameter of 15.03 mm. A 150 V, 40 A, 2 kW supply drives the selected element at the power ceiling. The 0D balance gives 1084.2 °C; the 2D field gives an average of 1096.4 °C and a peak of 1110.0 °C. The selected case remains below the 1200 °C display threshold.

3.2. Target-temperature capacity maps

The main and supplementary screening studies comprise 6,833 lumped evaluations of whether available electrical power can balance heat loss at the target temperature. Map colors show available power divided by required power, both evaluated at 1200 °C (**Figure 2**). A ratio above one indicates sufficient power to maintain the lumped target under the stated losses, with output reduced to the required level. A ratio below one identifies a power deficit; unity marks the capacity boundary. Startup time and internal peak temperature are evaluated separately.

The current follows eq 3 with zero contact resistance, giving available power I^2R; required power comes from the enclosure heat-loss network.

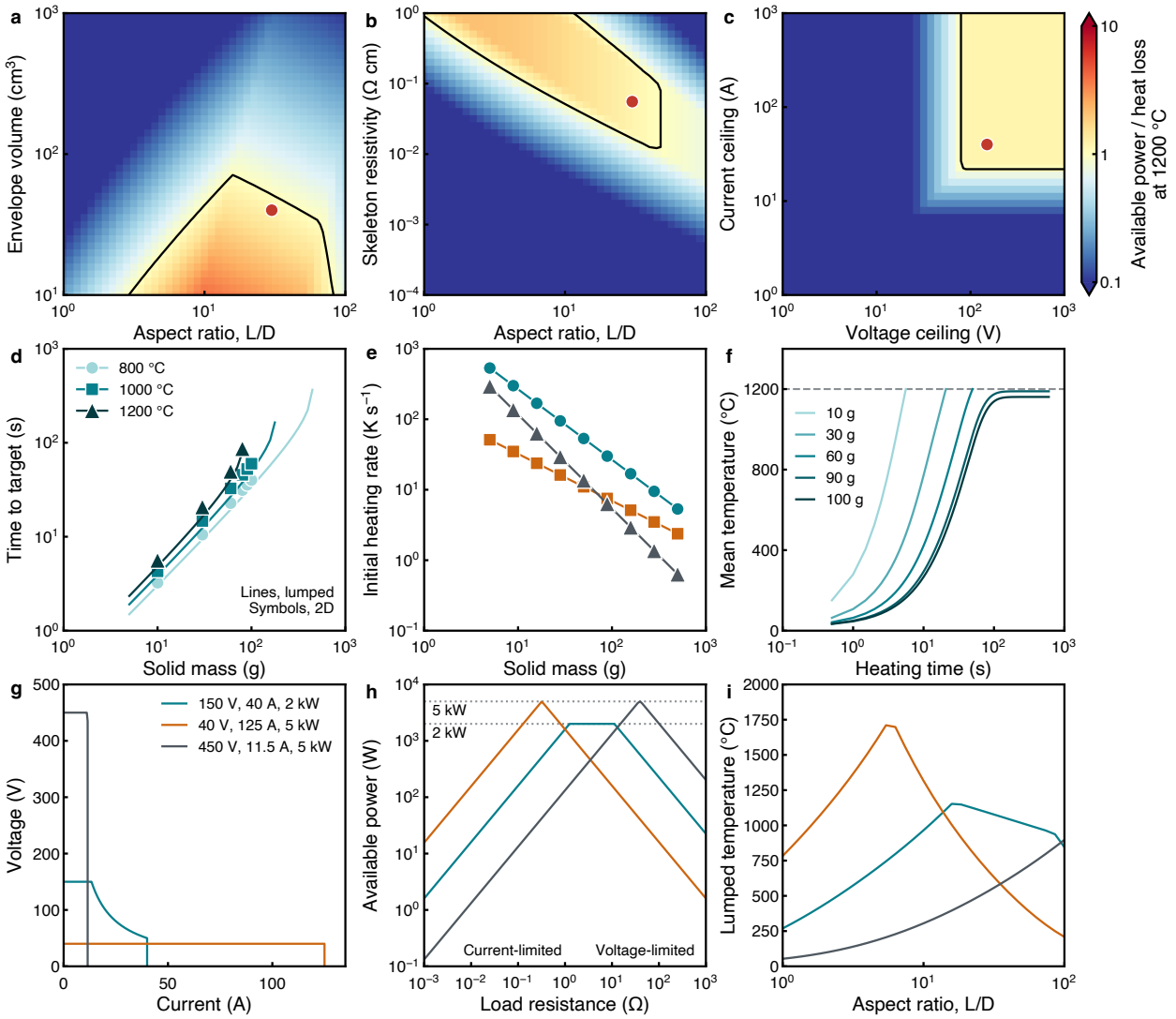


Figure 2. Heating capacity, startup and power-supply selection for porous SiC elements. **(a–c)** Available power divided by enclosure heat loss at 1200 °C versus **(a)** envelope volume and aspect ratio, **(b)** skeleton resistivity and aspect ratio at 40 cm^3, and **(c)** supply voltage and current ceilings at 40 cm^3 and L/D = 30. Black contours mark unity, and red points mark the 150 V, 40 A, 2 kW design. **(d)** Time to reach 800–1200 °C (lines lumped, symbols 2D) and **(e)** initial heating rate versus solid mass, and **(f)** 2D mean-temperature histories, at L/D = 30 and 50% porosity. Panels d and f use the 150 V, 40 A, 2 kW supply. **(g–i)** Three commercial supplies, colored as in e: **(g)** upper operating boundaries, **(h)** available power versus load resistance and **(i)** lumped temperature of an 80 cm^3 porous SiC element (solid fraction 0.5) versus L/D.

The capacity-map reference uses a smaller, 40 cm^3 envelope at L/D = 30. Its available 2.00 kW exceeds the required 1.71 kW by 17.3% of the required power. Its target-temperature resistance is 3.56 Ω. This example differs from the 80 cm^3 case used for the spatial grid study and transient calculation. Holding volume fixed in panel B separates the effect of resistivity and shape; holding the element fixed in panel C identifies the electrical specifications needed for the same target.

At low resistance the current ceiling limits heating; at high resistance the voltage ceiling limits it. When the power and current-density caps are inactive, the crossover is:

$$R(L^*) = \frac{V_{\max}}{I_{\max}}$$
$$L^* = \sqrt{\frac{V_{\max} \quad \mathcal{V}}{I_{\max} \quad \rho_e}} \tag{7}$$

This crossover marks the change in the active electrical constraint. Geometry also changes exposed area and enclosure losses, which affect the thermal optimum. In panel c, increasing both electrical ceilings eventually gives no further capacity because the power cap is fixed at 2 kW. The maps identify whether a design needs a change in resistance, supply capacity or heat loss to maintain the target under the stated porous-body approximation. Internal temperature distributions are examined separately in **Figure 3**. Figure S4 and Figure 2g–i extend the screening to other materials and commercial supplies.

Startup introduces a mass-dependent time scale (**Figure 2d–f**). The lumped heating time is obtained by integrating $mC_p/(P - Qloss)$ over temperature using the application's power and enclosure-loss functions. In panel d, the 0D lines use quasi-steady enclosure losses, whereas the 2D points include heat storage in the wall and surrounding domain; their heating times therefore differ. At fixed L/D and porosity, increasing solid mass enlarges both the element and its heat-loss area. The 2 kW supply heats the 10, 30 and 60 g cases to a mean 1200 °C in approximately 6, 21 and 50 s on the plotted grid. The 90 and 100 g cases remain below that target after 600 s. A larger rated supply can deliver less power to the same resistance: the 40 V / 125 A and 450 V / 11.5 A cases are voltage- and current-limited, respectively.

3.3. Material-dependent temperature distributions

To compare spatial temperature variation at the same lumped target, we select the current for each material to yield 900 °C in a 1.18 cm^3 dense cylinder in the 0D model. The same current setpoints are used in 2D, where the resolved heat balance can yield different average temperatures. The calculated maximum–minimum spread is 7.9 K for SiC, 20.7 K for SiSiC, 33.9 K for Inconel 601 and 68.4 K for FeCrAl (**Figure 3**). The resolved SiC peak lies 2.8 K above its average; the FeCrAl peak lies 24.7 K above its average. The axial and radial profiles locate these gradients within the element. Spatial temperature variation also affects catalytic use of Joule-heated supports: Burentugs and Lobo reported nonuniform surface temperatures in a catalyst-coated FeCrAl coil.[16] Malhotra et al. examined temperature homogeneity and solid–fluid equilibration in microwave-heated structured reactors.[17]

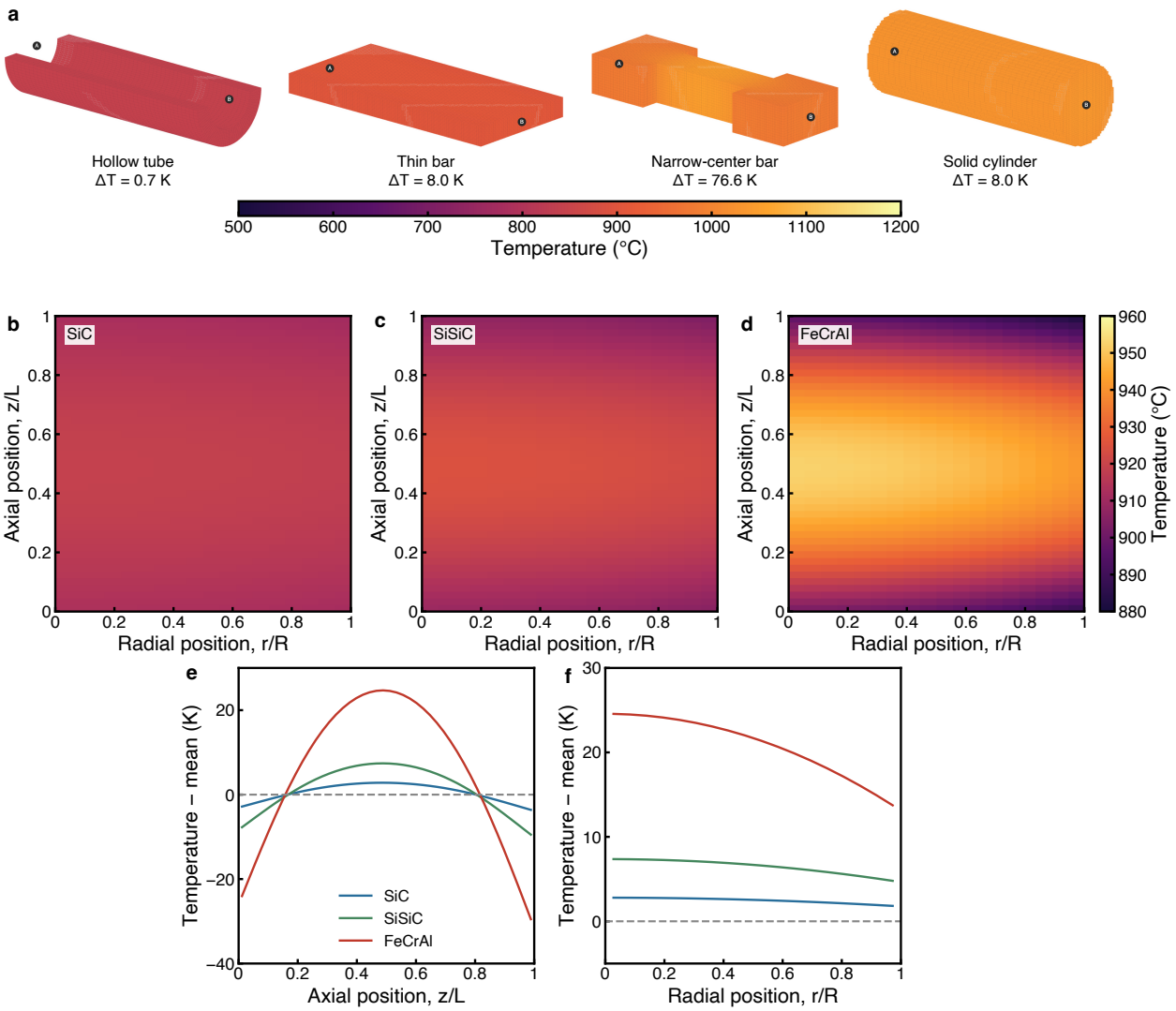


Figure 3. Temperature uniformity set by element geometry and material. **(a)** Steady temperature fields of four SiSiC elements of 1.0 cm^3 at 100 W, with the maximum minus minimum solid temperature ΔT. The hollow tube is cut away for display, and all values refer to the full bodies. Mean temperatures are 844, 912, 1003 and 1033 °C, and maxima are 844, 915, 1052 and 1037 °C, from left to right. **(b–f)** Dense SiC, SiSiC and FeCrAl cylinders (1.18 cm^3, L/D = 3), each at the current that gives a 900 °C lumped temperature: **(b–d)** solid temperature fields on one color scale, and **(e)** centerline axial and **(f)** midplane radial temperatures relative to the volume average.

FeCrAl conductivity is held at 11 W m^{-1} K^{-1} across the temperature range. Inconel retains its tabulated temperature dependence. Resistivity, emissivity and delivered power also differ. These compact dense elements form a separate case from the porous SiC design in **Figure 2**. Figure S2 examines the sensitivity of the porous design to thermal properties, enclosure losses and contact conditions.

3.4. Electrode contact width and supply limits

We next vary electrode contact width, which sets the contacted area on each end face, to examine how current distribution and supply limits affect heating. At 30 W, widening the electrode contact

regions reduces the calculated temperature spread from 31.7 to 13.5 K (**Figure 4**). The wider contacts lower the resistance and reach the 375 A ceiling at a lower power setpoint, despite the 15 kW power rating. At a 90 W setpoint, delivered power falls from 70.1 W for narrow contacts to 55.0 W for wide contacts. Once current is limited, the lower resistance reduces I^2R heating. Contact width and current capacity therefore determine both temperature uniformity and available heating power.

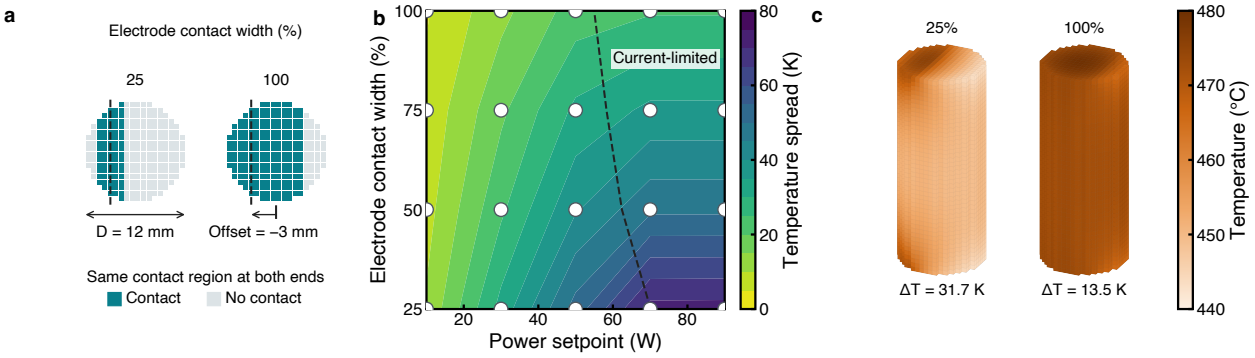


Figure 4. Electrode contact width and supply-limited heating. **(a)** Contact regions covering 25% and 100% of the 12 mm diameter, centered 3 mm off axis at both ends of a 30 mm cylinder. **(b)** Steady temperature spread versus power setpoint and contact width. White markers are calculations, and points right of the dashed $I^2_{\max}R$ line are current-limited. **(c)** Temperature fields at 30 W for the two contact widths. The supply limits are 40 V, 375 A and 15 kW, and the element has resistivity 1.45×10^{-6} Ω m and conductivity 11 W m^{-1} K^{-1}.

For Figure 4, exposed non-electrode surfaces exchange heat with a 25 °C ambient through h = 15 W m^{-2} K^{-1} and emissivity 0.8. Electrode heat loss is zero; gas flow, enclosure heat transfer and process heat duty are disabled.

3.5. Element geometry

At fixed power, geometry changes the exposed surface area and heat-loss paths, shifting the steady temperature and its distribution. For SiSiC elements of approximately 1 cm^3 solid volume at 100 W, the calculated mean temperature ranges from 844 °C for the hollow tube to 1033 °C for the solid cylinder (**Figure 3a**). The narrow-center bar reaches the highest local temperature, 1052 °C, with a 76.6 K temperature spread; the reduced cross-section concentrates current and restricts axial heat conduction. Figure S5 gives the FeCrAl comparison with a 1000 A current ceiling.

The ScreenJoule SiSiC preset supplies tabulated resistivity and thermal conductivity, with density 3050 kg m^{-3} and heat capacity 680 J kg^{-1} K^{-1}; the preset property tables and dimensions are recorded in the calculation settings. Every exposed surface, including the tube bore, exchanges heat with a 25 °C ambient through h = 15 W m^{-2} K^{-1} and emissivity 0.8. Full-end ideal electrodes have zero contact resistance and zero electrode heat loss. Supply limits are 150 V, 40 A and 2 kW; all cases deliver 100 W within these limits and the prescribed 5×10^6 A m^{-2} current-density ceiling. The tube uses a 12×96×48 annular mesh; the bars use 40×16×90 and the cylinder 40×40×36 Cartesian meshes.

4. Conclusions

Supply limits and thermal losses jointly determine whether an element reaches its target temperature. Wider electrode contacts improve uniformity at fixed power while reducing available

heating under a current ceiling. At fixed power and similar solid volume, element shape changes mean temperature and local hot spots through exposed area, current concentration, and conduction paths. Users can extend the material models, geometries and boundary conditions and check their modifications against the verification benchmarks.

Associated Content

Supporting Information contains geometry and property definitions, finite-volume equations, numerical diagnostics, boundary sensitivities, material-specific maps and supply comparisons. The public repository is available at https://github.com/robin-yk/ScreenJoule.

Acknowledgments

Generative AI tools assisted with website development, implementation of the three-dimensional flow and electrothermal models, and preparation of documentation and figures. The authors reviewed the code and verified the reported calculations.

References

1. Mittal, A.; Ierapetritou, M.; Vlachos, D. G. Joule-heating materials for advanced chemical manufacturing. *Chem. Eng. J.* **2025**, *520*, 166348.

2. Kim, Y. T.; Lee, J. Electricity-driven reactors that promote thermochemical catalytic reactions via Joule and induction heating. *Chem. Eng. J.* **2023**, *470*, 144333.

3. Wismann, S. T.; Engbæk, J. S.; Vendelbo, S. B.; Bendixen, F. B.; Eriksen, W. L.; Aasberg-Petersen, K.; Frandsen, C.; Chorkendorff, I.; Mortensen, P. M. Electrified methane reforming: A compact approach to greener industrial hydrogen production. *Science* **2019**, *364*, 756–759.

4. Zheng, L.; Ambrosetti, M.; Marangoni, D.; Beretta, A.; Groppi, G.; Tronconi, E. Electrified methane steam reforming on a washcoated SiSiC foam for low-carbon hydrogen production. *AIChE J.* **2023**, *69*, e17620.

5. Nicolini, F.; Ambrosetti, M.; Beretta, A.; Groppi, G.; Tronconi, E. Directly electrified SiSiC packed foam reactor for methane steam reforming: Experimental study and model-based scaleup. *Int. J. Hydrogen Energy* **2026**, *206*, 153393.

6. Kwak, Y.; Railkar, R.; Zheng, W.; Vlachos, D. G. Tandem nonoxidative coupling of methane and carbon dioxide reduction via pulsed Joule thermochemistry. *ACS Energy Lett.* **2025**, *10*, 6188–6196.

7. Mittal, A.; Kwak, Y.; Zheng, W.; Ierapetritou, M.; Vlachos, D. G. Short contact time, high temperature, internally heated ethane crackers. *Chem. Eng. J.* **2025**, *523*, 168251.

8. Mittal, A.; Ierapetritou, M.; Vlachos, D. G. Computational insights into steady-state and dynamic Joule-heated reactors. *React. Chem. Eng.* **2024**, *9*, 2380–2392.

9. Sundaramoorthy, A. S.; Lobo, R. F.; Vlachos, D. G. Coarse-grained models for scale-up of structured reactors. *Ind. Eng. Chem. Res.* **2025**, *64*, 14110. DOI: 10.1021/acs.iecr.5c01707.

10. Xie, C. Interactive Heat Transfer Simulations for Everyone. *The Physics Teacher* **2012**, *50* (4), 237–240.

11. Garg, S.; Fornaciari, J. C.; Weber, A. Z.; Danilovic, N. fuelcell: A Python Package and Graphical User Interface for Electrochemical Data Analysis. *J. Open Source Softw.* **2021**, *6* (59), 2940. DOI: 10.21105/joss.02940.

12. Medasani, B.; Kasiraju, S.; Vlachos, D. G. OpenMKM: An open-source C++ multiscale modeling simulator for homogeneous and heterogeneous catalytic reactions. *J. Chem. Inf. Model.* **2023**, *63*, 3377–3391.

13. Patankar, S. V. *Numerical Heat Transfer and Fluid Flow*; Hemisphere: Washington, DC, 1980.

14. van der Vorst, H. A. Bi-CGSTAB: A fast and smoothly converging variant of Bi-CG for the solution of nonsymmetric linear systems. *SIAM J. Sci. Stat. Comput.* **1992**, *13*, 631–644.

15. Oberkampf, W. L.; Roy, C. J. *Verification and Validation in Scientific Computing*; Cambridge University Press: Cambridge, 2010.

16. Burentugs, E.; Lobo, R. F. Catalytic Ethane Dehydrogenation Using a Porcupine Heating Element in a Joule-Heated Reactor. *Ind. Eng. Chem. Res.* **2025**, *64* (19), 9835. DOI: 10.1021/acs.iecr.5c00479.

17. Malhotra, A.; Chen, W.; Goyal, H.; Plaza-González, P. J.; Julián, I.; Catalá-Civera, J. M.; Vlachos, D. G. Temperature Homogeneity under Selective and Localized Microwave Heating in Structured Flow Reactors. *Ind. Eng. Chem. Res.* **2021**, *60* (18), 6835. DOI: 10.1021/acs.iecr.0c05580.

Supporting Information

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Yeonsu Kwak*

Department of Chemical and Biomolecular Engineering,
University of Delaware, Newark, Delaware 19716, United States

*E-mail: fiske@udel.edu

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S1. Numerical methods

Sections S1.1–S1.4 describe the lumped and axisymmetric formulations. Section S1.5 gives the three-dimensional implementation details.

Table S1. Formulations and gas-flow treatment in the ScreenJoule workspaces. The entries describe the implemented models; settings for each reported calculation are specified with that calculation.

Quantity	0D	2D axisymmetric	3D
Solid temperature	Lumped energy balance	Radial and axial conduction	Three-dimensional conduction
Electrical heating	Bulk resistance and supply limits	Potential field and local Joule heating	Potential field and local Joule heating
Gas energy	Enclosure heat-exchange approximation for outlet temperature	Advection and conduction in gas cells	Advection and conduction in gas cells, coupled to the solid
Gas motion	No spatial velocity or pressure field	Prescribed axial flow; no momentum or pressure solve	Incompressible Stokes velocity and pressure; fully developed branch for invariant thin-plate cross-sections
Gas inputs	Enclosure He-flow preset	He at 50 sccm in the current purge implementation	User-specified flow rate and gas properties
Principal outputs	Operating power, representative temperature and heating time	Temperature and potential fields; gas outlet temperature	Temperature and potential fields; gas velocity, pressure drop and outlet temperature

Cross-dimensional comparisons use matched geometry, properties, power and thermal boundaries. The 2D gas-energy calculation uses a prescribed flow distribution, whereas the 3D gas mode obtains velocity from the Stokes equations. The Stokes formulation omits convective inertia. The shared electrothermal benchmarks use gas flow disabled.

S1.1. Domain and boundaries

The reference SiC element has an envelope volume of 80 cm^3 and a void fraction of 0.50, surrounded by a 0.5 mm gap and a 1 mm quartz wall. Ambient and inlet temperatures are 20 °C, with external h = 12 W m^{-2} K^{-1}. Steady and transient calculations use 30 × 60 and 20 × 40 meshes, respectively.

The axisymmetric domain comprises the r-z half-plane. The radial boundary is the symmetry axis $r = 0$. The outer radial face is fixed at the ambient temperature through the half-cell conductive resistance. At the two axial domain ends, nonflowing cells use the same fixed-ambient condition. Flow-connected gas cells use the prescribed inlet enthalpy at the top and advective outflow with zero axial conductive flux at the bottom. The element ends couple to the gas pads by conduction and radiation.

S1.2. Geometry and properties

$$\begin{aligned} V_{env} &= \frac{V_s}{f_s} \\ D &= \left(\frac{4V_{env}}{\pi\lambda}\right)^{1/3} \\ L &= \lambda D \\ A_{eff} &= f_s \frac{\pi D^2}{4} \\ A_s &= \pi DL + \frac{\pi D^2}{2} \end{aligned} \tag{S1}$$

In the porous-body approximation, the solid fraction determines the conducting area and thermal storage. An independently specified isotropic effective thermal conductivity describes heat conduction. Properties specified per bulk volume are used without a second porosity correction. For a rectangular element of length L, width W, and thickness H, $V_{env} = LWH$, $A_{eff} = f_s WH$, and $S = 2(LW + LH + WH)$. The zero-dimensional calculation uses the rectangular dimensions; the axisymmetric solver substitutes a cylinder with the same length and radiating surface, with effective density preserving mass and effective resistivity preserving resistance (Figure 1e).

The lumped and axisymmetric solvers accept tabulated properties, tabulated resistivity multipliers, a linear temperature coefficient, or constant values. Outside a tabulated range, they retain the nearest endpoint value. Their linear resistivity expression has a minimum of 5% of its room-temperature value,

$$\rho_e(T) = \rho_0 \max[0.05, \quad 1 + \alpha_\rho(T_{\circ C} - 20)] \tag{S2}$$

The zero-dimensional model solves $P_{bulk}(T) = Q_{loss}(T)$ by bracketing and bisection. In the direct-surface limit $Q_{loss} = \varepsilon\sigma_{SB}A_s(T^4 - T_a^4) + hA_s(T - T_g)$; the enclosure model replaces this with element-to-wall conduction and radiation, wall conduction, outside-air conduction, external convection, boundary radiation, and axial losses. The zero-dimensional result initializes the 2D iteration and supplies the lumped comparisons in Figure 2 and the associated Supporting Information.

S1.3. Finite-volume discretization

For a finite volume bounded by radial coordinates r_w, r_e and axial edges z_s, z_n,

$$
\begin{aligned}
V_p &= \pi(r_e^2 - r_w^2)(z_n - z_s) \\
A_w &= 2\pi r_w \Delta z \\
A_e &= 2\pi r_e \Delta z \\
A_s = A_n &= \pi(r_e^2 - r_w^2)
\end{aligned}
\tag{S3}
$$

The thermal conductance between adjacent cells p and q is obtained from the sum of the two half-cell thermal resistances,

$$
G_{pq} = \frac{A_f}{d_p/k_p + d_q/k_q} \tag{S4}
$$

The resulting intercell fluxes enter the neighboring energy balances with opposite signs. The linearized radiative heat-transfer coefficient is $h_{rad} = \varepsilon\sigma_{SB}(T + T_s)(T^2 + T_s^2)$; for concentric element and wall surfaces,

$$
\varepsilon_{eff} = [\frac{1}{\varepsilon_e} + \frac{R_e}{R_w}(\frac{1}{\varepsilon_w} - 1)]^{-1} \tag{S5}
$$

Element-to-wall radiation is evaluated between nonadjacent cells in parallel with conduction through the gap. External convection acts at wall–air interfaces, wall radiation exchanges heat with the prescribed ambient, and the surrounding-air cells conduct heat to the outer boundary. At a helium flow rate of 50 sccm, the heat-capacity rate is $\dot{C}_{He} = 7.73 \times 10^{-4}$ W K^{-1}; each downstream gas row receives enthalpy from the corresponding upstream cell.

S1.4. Field solution

The electrical potential is calculated within the element using terminal potentials of 0 and 1 V and zero normal current at the curved surface. Face electrical conductances follow eq S4 with thermal conductivity replaced by electrical conductivity. The face dissipation contributions are summed for each cell and normalized to P_{bulk}. The field-derived resistance is reported.

The lumped steady temperature initializes the field on a fixed mesh. Each Picard iteration evaluates T_{avg} and updates scalar resistivity, resistance, current and P_{bulk}. With the local electrical calculation enabled, the solver updates the unit-potential field and heat-source distribution. It updates thermal conductivity, heat capacity, gas enthalpy and radiation coefficients, solves the thermal matrix, and under-relaxes the field using $T^{n+1} = T^n + \omega(T^* - T^n)$, with ω initially set to 0.62. The steady iteration stops when the temperature update meets the specified tolerance. Linear residual and energy closure are reported separately. Jacobi preconditioning scales each row by its diagonal. Preconditioned conjugate gradients (PCG) solve the symmetric operator; BiCGSTAB handles the nonsymmetric operator produced by directed gas transport, both to a relative tolerance of 10^{-11}. Backward Euler adds $\rho c_p V/\Delta t$ to the diagonal with the heat capacity evaluated at the midpoint of the step. The closure metric is

$$
\varepsilon_E = \frac{|P_{bulk} - Q_{boundary} - \dot{E}_{stored}|}{\max(P_{bulk}, Q_{boundary}, |\dot{E}_{stored}|, 10^{-12})} \tag{S6}
$$

Stored energy is obtained by integrating heat capacity over temperature and accounting for the porous solid fraction. When a step crosses a property-table knot, the discrete energy balance and the integrated stored-energy change can differ because they use different heat-capacity evaluations.

S1.5. Three-dimensional implementation

The 3D solver uses finite-volume electrical and thermal conductances on Cartesian or annular meshes. It evaluates material properties at each cell temperature and rejects temperatures outside the supplied property tables. Its linear resistivity law uses a 25 °C reference and requires positive resistivity. Electrical and thermal solutions are updated in damped Picard iterations. Contact dissipation is shared between adjacent material cells; dissipation at an ideal terminal enters the neighboring solid cell.

S1.6. Equivalent insulation screening

The 0D insulation tool calculates the equivalent layer thickness needed to reach a specified temperature at fixed delivered heater power. User-specified process and gas heat duties are subtracted from this power. The remaining heat flux determines the outer-surface temperature from convection and radiation; the temperature drop across the layer then determines its resistance and thickness for an assigned thermal conductivity. The calculation uses the element envelope area on all faces and holds this area constant. Measured power and temperature can also be used to estimate an equivalent system resistance, which includes heat losses not separately specified. A thickness sweep starts from the exposed case at zero thickness; electrical supply limits are evaluated in the main model.

S2. Numerical verification

The manufactured-solution test gives a finest-mesh L2 error of 0.0223 K. The largest annular discrepancy at L/D = 100 is 1.95%; reducing the time step from 2.5 to 1.25 s changes the endpoint temperature by 1.327 K. All 112 regression tests pass.

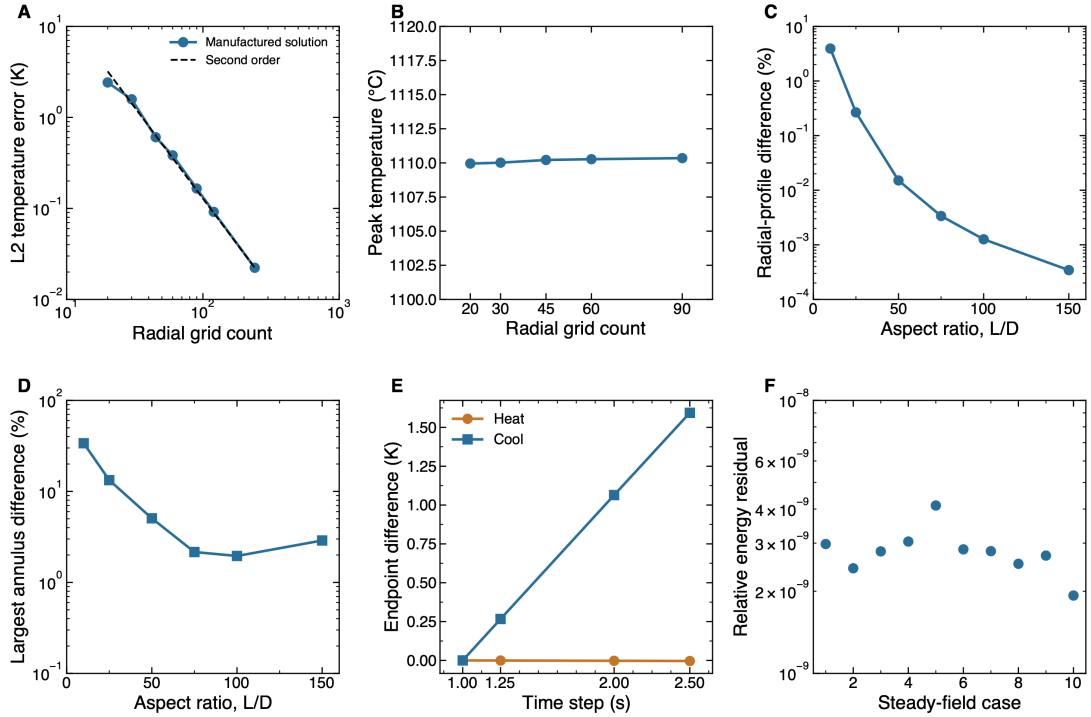


Figure S1. Numerical verification. **(A)** Manufactured-solution errors on seven grids with

20, 30, 45, 60, 90, 120 and 240 radial cells and twice as many axial cells. The dashed line gives second-order scaling anchored to the finest grid. **(B)** Porous SiC peak temperature on five meshes with 20, 30, 45, 60 and 90 radial cells. The two finest meshes differ by 0.079 K. **(C,D)** Finite-cylinder calculations at L/D = 10, 25, 50, 75, 100 and 150 compared with radial-parabola and annular-conduction solutions for infinitely long cylinders. Deviations include finite-end effects. **(E)** End-of-leg mean-temperature differences for time steps of 1, 1.25, 2 and 2.5 s, relative to the 1 s result for each leg. Each heating leg starts at ambient and lasts 300 s; its 300 s cooling leg starts from the corresponding heated field. **(F)** Relative energy residuals for the original ten steady-field cases. Symbols show calculations; connecting lines guide the eye. Energy-balance closure and temporal convergence are separate checks.

S3. Design sensitivities

S3.1. Material comparison

Table S2. Inputs and calculated operating points for Figure 3. Values use the 30 × 60 mesh. Ceramic cases use 150 V / 40 A / 2 kW ceilings; alloy cases use 40 V / 375 A / 15 kW ceilings, with current setpoints below those limits. Contacts are ideal and lead losses are omitted; low terminal voltage makes the alloy cases sensitive to those omissions.

Material	k at 2D average (W m^{-1} K^{-1})	Current (A)	Voltage (V)	Bulk power (W)	2D average / peak (°C)	Preset source / thermal model
SiC	120.0	13.85	3.701	51.27	916.1 / 918.9	Mittal et al. (2025), Table 1; Constant k proxy
SiSiC	48.1	26.64	2.028	54.03	917.1 / 924.5	550-750 °C bulk ρ back-calculated from Zheng et al., AIChE J. 69, e17620 (2023), Table 2; RT branch is a commercial SiC-element proxy; Tabulated k(T)
FeCrAl	11.0	258.15	0.187	48.17	927.6 / 952.3	Kanthal resistance materials handbook; Constant k proxy
Inconel 601	26.5	290.97	0.176	51.30	921.2 / 933.2	Special Metals Inconel 601 bulletin, Table 3; Tabulated k(T)

Refinement from 30 × 60 to 45 × 90 cells changes the SiC and FeCrAl temperature extrema by at most 1.43 K. All four material cases converge with relative energy residuals below 6×10^{-10}. $MoSi_2$ is excluded because the selected geometry cannot reach the matched target within its preset current-density limit.

S3.2. Thermal properties and contacts

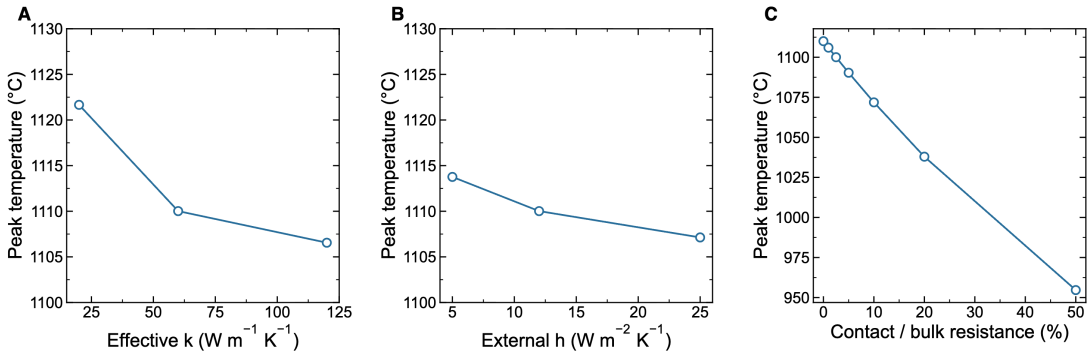


Figure S2. Sensitivity of the steady-state peak element temperature for the selected porous SiC geometry. (A) Effective thermal conductivity. **(B)** External convection coefficient. **(C)** Combined resistance of both electrical contacts divided by the bulk element resistance evaluated at 1200 °C.

Contact resistance is prescribed for the sensitivity sweep and held constant with temperature; bulk resistance varies with temperature.

For Figure S6d, the combined contact resistance spans 0–50% of the bulk resistance at 1200 °C (2.8238 Ω), corresponding to 0–1.4119 Ω. Interface resistivity equals half the combined contact resistance multiplied by the model current-carrying area.

S3.3. Element dimensions

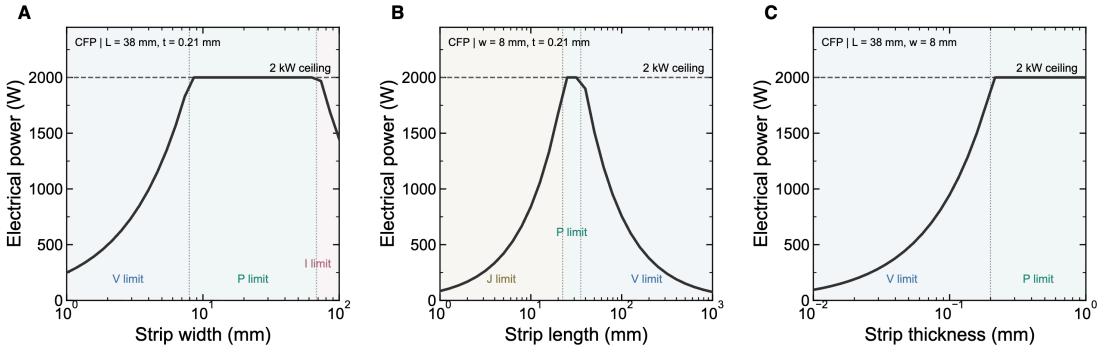


Figure S3. Electrical constraints during dimensional sweeps under 150 V, 40 A and 2 kW supply ceilings and the preset current-density limit. Shaded regions identify the active constraint at each calculated operating point; dotted boundaries are placed between adjacent samples. I, J, P and V denote current, current-density, power and voltage limits, respectively. **(A)** CFP width at fixed length and thickness. Increasing width lowers resistance, shifting operation from voltage limitation through the power ceiling to current limitation. **(B)** CFP length at fixed cross-section. Power increases under the current-density limit, reaches the power ceiling and decreases under the voltage limit. **(C)** CFP thickness at fixed length and width. Increasing thickness reduces resistance until the power ceiling is reached. All panels report electrical power. Dimensional changes also affect mass and exposed area, so temperature requires a thermal calculation. The electrical comparisons include cases above 1200 °C.

S3.4. Materials and supplies

The material calculations use the specified preset properties and material-dependent geometry ranges. Power-supply envelopes are constructed from the nominal ratings supplied by the author.

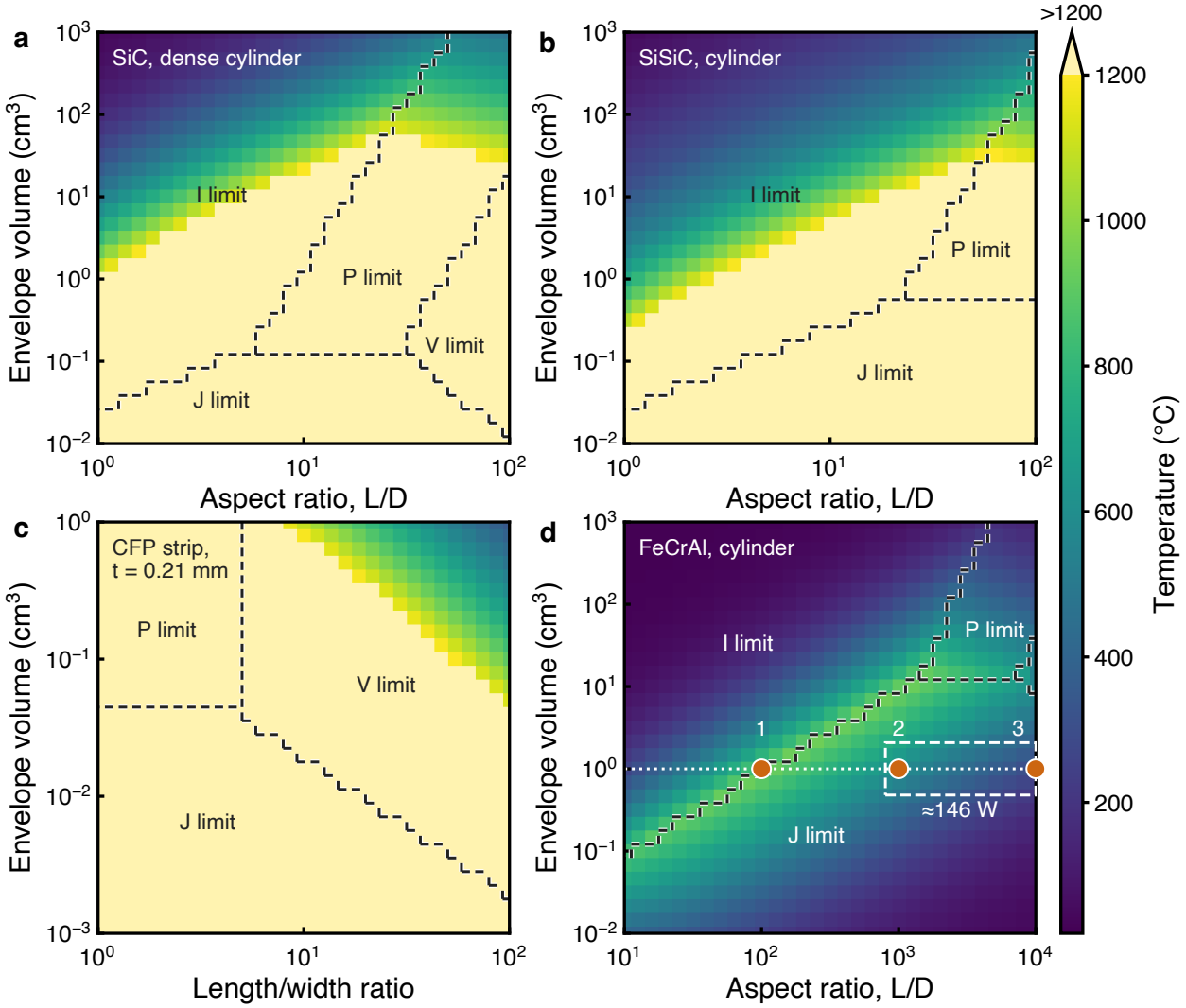


Figure S4. Lumped material screening under 150 V, 40 A and 2 kW limits. **(a)** Dense SiC, **(b)** SiSiC and **(d)** FeCrAl cylinders and **(c)** a 0.21 mm CFP strip. Pale yellow marks temperatures above 1200 °C, and dashed lines separate the active current (I), current-density (J), power (P) and voltage (V) limits. Points 1–3 in d mark 1 cm^3 FeCrAl cylinders at L/D = 100, 1000 and 10,000. The current-density limit holds the power near 146 W from point 2 to point 3.

In panel d, the 1 cm^3 cylinders become cooler as L/D increases from 1000 to 10000: the current-density limit holds power near 146 W while exposed area increases. These calculations treat straight cylinders.

S3.5. FeCrAl geometries

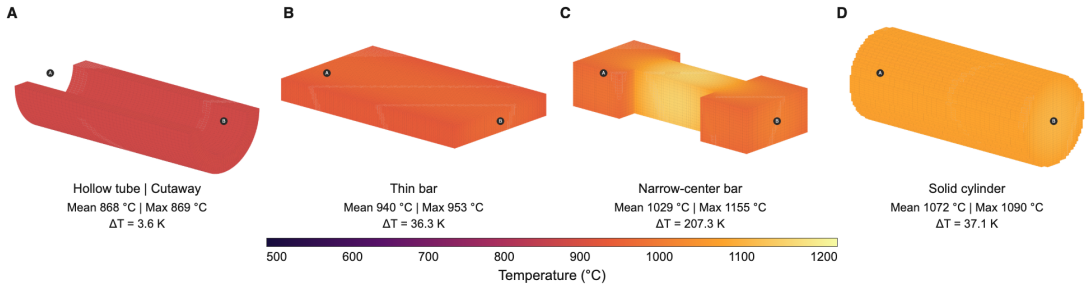


Figure S5. Steady temperature fields for four FeCrAl geometries at 100 W: (A) hollow tube, **(B)** thin rectangular bar, **(C)** bar with a narrow center, and **(D)** solid cylinder. Panels are ordered by volume-average temperature and share the same viewing angle and 500–1200 °C color scale. Panel A uses a longitudinal half-cutaway; all reported temperatures refer to the complete body. ΔT is the maximum minus minimum solid temperature.

The FeCrAl elements have solid volumes of 1.000–1.021 cm^3, resistivity 1.45×10^{-6} Ω m, and thermal conductivity 11 W m^{-1} K^{-1}. All exposed surfaces, including the bore, exchange heat with a 25 °C ambient through h = 15 W m^{-2} K^{-1} and emissivity 0.8. Ideal full-end electrodes contribute no contact resistance or heat loss. The 150 V / 1000 A / 2 kW limits permit 100 W in every geometry.

S4. Literature comparisons

Figure S6 compares three thermal-boundary calculations: a cylindrical thermal-resistance calculation for the FeCrAl tube (Wismann), a 3D calculation with prescribed process heat duty for the SiSiC foam (Zheng), and a lumped exposed-surface calculation for carbon fiber paper (Mittal and Kwak).

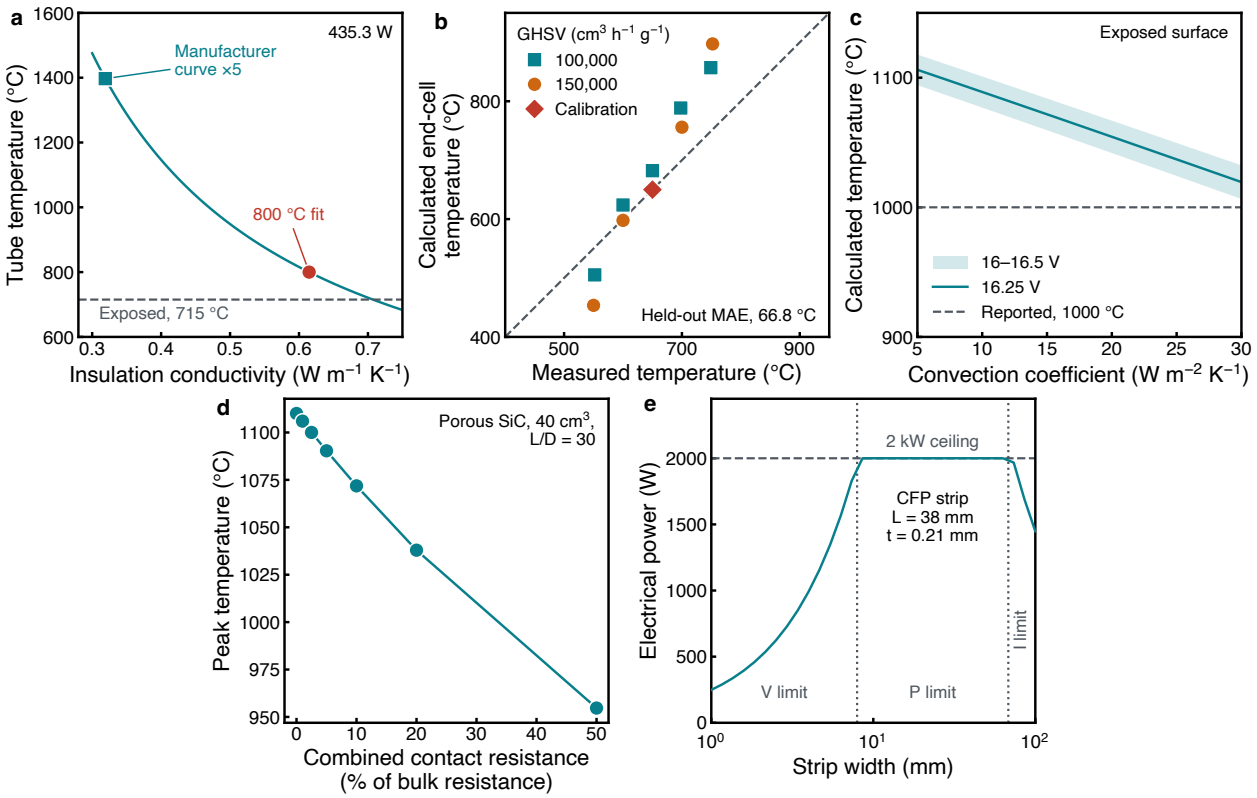


Figure S6. Model calibration and design sensitivity. **(a)** Wismann FeCrAl tube temperature at 435.3 W versus equivalent insulation conductivity. The red point marks the conductivity that gives the reported 800 °C, and the square uses a manufacturer conductivity curve multiplied by five. **(b)** Calculated end-cell versus measured downstream temperatures for ten Zheng operating points at two space velocities. **(c)** Exposed-surface CFP temperature versus external convection coefficient at 16–16.5 V. The dashed line marks the reported 1000 °C. **(d)** Peak temperature of the porous SiC design versus combined contact resistance. **(e)** Electrical power of a 38 mm long, 0.21 mm thick CFP strip versus width, with the active electrical limit marked.

FeCrAl tube (Wismann). The thermal-resistance calculation balances an assumed input of 435.3 W against insulation, external-surface and end losses, with zero reaction and gas enthalpy loads. Equivalent insulation conductivity is fitted to 800 °C.

SiSiC foam (Zheng). Reported power and enthalpy duty are inputs. A surface heat-loss coefficient of 17.05 W m^{-2} K^{-1} is fitted at 650 °C and GHSV = 150,000 cm^3 h^{-1} $gcat^{-1}$, then retained for the other nine points (mean absolute error 66.8 °C). End-cell temperature represents the downstream measurement; the contactor and thermocouple are not resolved.

Carbon fiber paper (Mittal and Kwak). The exposed CFP strip is represented by an equivalent cylinder at ambient 20 °C and emissivity 0.57. Gas enthalpy and clamp conduction are omitted. The voltage interval represents the reported operating range.

S5. OpenFOAM comparison of gas flow and conjugate heat transfer

Six cases compare ScreenJoule with an independently assembled OpenFOAM 2512 finite-volume implementation of fully developed flow and three-dimensional conjugate heat transfer (Figure S7). A $40 \times 10 \times 2$ mm solid lies at the centre of a $40 \times 20 \times 6$ mm channel. Actual inlet flow rates are 1, 10 and 100 $cm^3\ min^{-1}$ on $12 \times 9 \times 24$ and $24 \times 9 \times 48$ channel meshes. The solid receives 1 W, has thermal conductivity 20 $W\ m^{-1}\ K^{-1}$, and exchanges heat at both ends with a 25 °C sink through a coefficient of 200 $W\ m^{-2}\ K^{-1}$. Gas enters at 25 °C; the outer channel sidewalls are held at 25 °C. Gas density, heat capacity, conductivity and viscosity are 1.2 $kg\ m^{-3}$, 1005 $J\ kg^{-1}\ K^{-1}$, 0.026 $W\ m^{-1}\ K^{-1}$ and 1.8×10^{-5} Pa s, respectively. Radiation and reactions are omitted.

For the invariant cross-section, both implementations solve the transverse viscous balance for axial velocity, with no slip at the heater and outer walls. OpenFOAM first solves at unit pressure drop and scales velocity and pressure drop to the specified flow. The axial pressure profile follows from this gradient. This comparison tests the fully developed branch. The thermal calculation uses harmonic face conductivities, first-order upwind gas advection, conductive and advective inlet terms, and zero conductive flux at the gas outlet. The constant-conductivity, full-electrode ScreenJoule case produces uniform heating; OpenFOAM applies the corresponding 1 W volumetric source.

Across the six matched cases, the maximum axial-velocity difference is 1.56×10^{-12} $m\ s^{-1}$ and the relative pressure-drop difference is below 2.70×10^{-14}. Maximum solid and gas temperature differences are 1.26×10^{-6} and 1.05×10^{-6} K, respectively. The flow-weighted outlet-temperature difference remains below 3.21×10^{-7} K. OpenFOAM closes the total energy balance within 2.20×10^{-13} W. These differences measure agreement between independent implementations of matching discrete equations. Refinement in the transverse width and axial directions changes the solid mean by approximately 1 K, the gas outlet by 0.45 K and the pressure drop by 9.4%; the channel-height resolution remains nine cells.

At 100 $cm^3\ min^{-1}$ on the finer mesh, the solid mean is 70.800 °C, the gas outlet is 37.248 °C and the pressure drop is 0.013029 Pa. The 1 W input leaves through the channel walls (0.599625 W), solid end contacts (0.342297 W), inlet conduction (0.033459 W) and outlet sensible heat (0.024619 W). Heat loss to the cold channel walls maintains a substantial solid–gas temperature difference across the tested flow range.

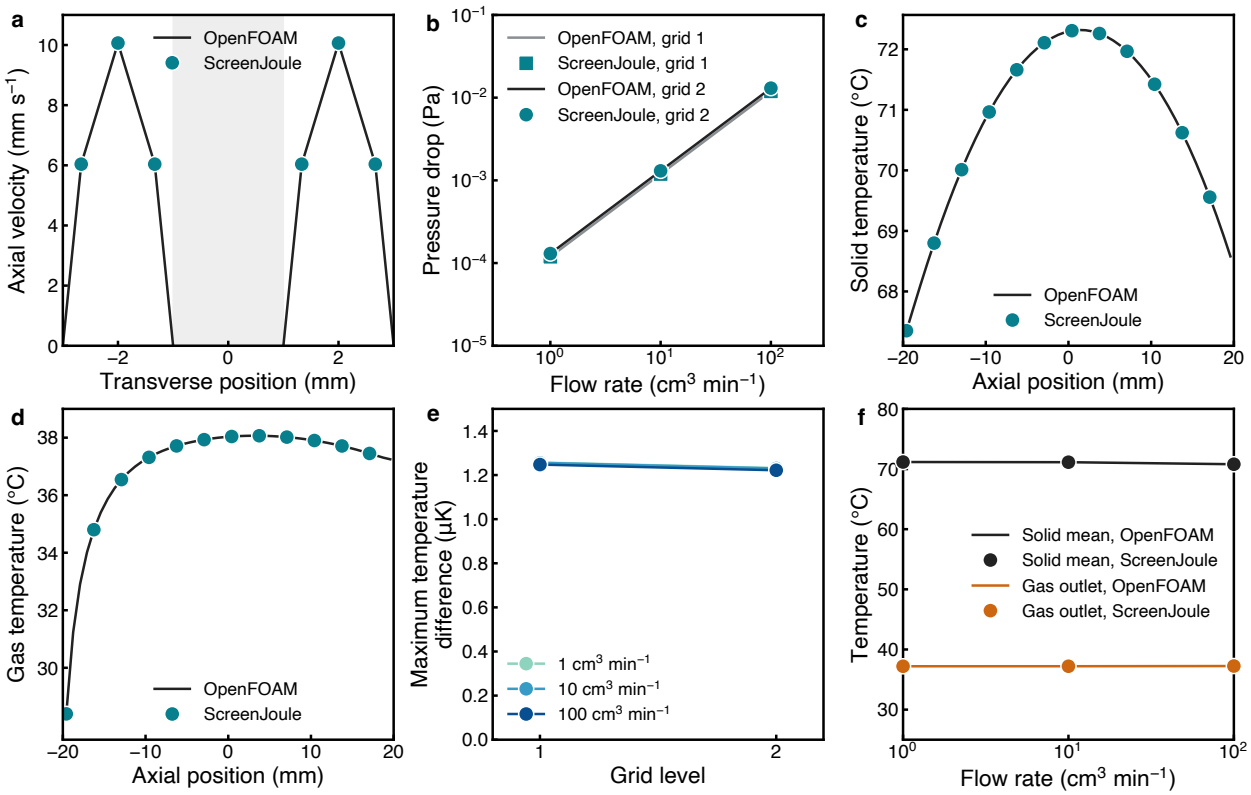


Figure S7. ScreenJoule and OpenFOAM for fully developed channel flow with conjugate heat transfer. ScreenJoule (teal symbols) and an independent OpenFOAM implementation (black lines) of a 1 W heated solid in a gas channel. **(a)** Axial velocity across the channel height, with the solid shaded. **(b)** Pressure drop versus flow rate on both grids. **(c)** Solid section-mean and **(d)** flow-weighted gas temperatures along the channel. **(e)** Maximum cell-temperature difference between the codes. **(f)** Solid mean and gas outlet temperatures versus flow rate. Panels a, c and d use 100 cm^3 min^{-1}, and panels a, c, d and f use grid level 2 ($24 \times 9 \times 48$). Grid level 1 is $12 \times 9 \times 24$.

Nomenclature

Symbols for the lumped and axisymmetric formulations.

Symbol	Meaning	Units
r, z	radial and axial coordinates; $r = 0$ is the symmetry axis	m
$T(r,z)$, T_{avg}	temperature field; volume average over the element	K
$\rho_e(T)$, $\sigma(T)$	electrical resistivity and conductivity	Ω m, S m^{-1}
Φ	electric potential in the unit-potential problem	V
L, D, λ	element length, equivalent diameter, aspect ratio L/D	m, m, –
V_s, $\mathcal{V}$, A_{eff}	conducting solid volume $A_{eff}L$; effective conducting area	m^3, m^2
R_{bulk}, R_c, R_{total}	element resistance; one contact; $R_{bulk} + 2R_c$	Ω

Symbol	Meaning	Units
I, $J_{\max}$	operating current; ceiling on I/A_{eff}	A, A m^{-2}
P_{bulk}, $P_{contact}$	I^2R_{bulk} deposited; $I^2(2R_c)$ reported	W
q''', q'''_{unit}	deposited source; unnormalized dissipation at 1 V	W m^{-3}
$k(T)$, $c_p(T)$	thermal conductivity; specific heat	W m^{-1} K^{-1}, J kg^{-1} K^{-1}
h, h_{rad}, h_{eff}	convection, linearized radiation, and combined surface coefficient	W m^{-2} K^{-1}
τ	lumped time constant $mc_p/(h_{eff}S)$	s